Assessing the Maturity of Sustainability-aware Data Management Strategies that minimise Data Volumes and reduce Energy Consumption: A Study Approach

Engelbert Fellinger [1] and Patricia Lago [2]

Abstract: Digital sustainability can be considered along four core dimensions: technical, social, economic, and environmental. We found a significant gap in research on assessing data management (DM) strategies aimed at minimising data volumes and reducing energy consumption. In this paper, we present a mixed-method study designed to create the SusDMMA-DVE model. Aimed to assess the maturity and effectiveness of sustainability-aware DM strategies, this model can help identify DM challenges, such as excessive data collection, thereby reducing energy consumption and operational costs while meeting business needs.

Keywords: Data management strategies, sustainability, energy efficiency, carbon dioxide emissions, data growth, data management maturity assessment

1 Introduction

Data is a valuable resource that provides insight and informs decision-making. However, excessive data collection can be counterproductive as it hinders business processes, consumes large amounts of energy, and increases costs for data processing and storage. As more data is generated at an exponential rate, concerns about growing energy demands and carbon dioxide emissions are increasing [Ca24]. According to research commissioned by Veritas, only 15% of organisational data is business-critical [Ve16]. Research by Netapp states that a significant two-fifths (41%) of the data stored by UK organisations is unused or unwanted, leading to unnecessary costs and emissions from storage and energy consumption [Wa23]. Data volumes can be minimised by applying sustainability-aware data management (DM) strategies such as compression, deduplication, and data classification that have different characteristics in terms of use and potential energy-savings [Fe26; FL25].

We define *sustainability-aware data management maturity* as the degree to which an organisation manages and improves its data processes to address sustainability goals, achieving both effectiveness and efficiency; this involves ensuring that data practices are not only efficient, cost-effective, legally compliant, secure, and measurable, but also minimise environmental impact, conserve resources, and support overall organisational sustainability goals.

[1] Vrije Universiteit Amsterdam, De Boelelaan 1111, 1081 HV Amsterdam, The Netherlands, e.h.fellinger@vu.nl, https://orcid.org/0009-0001-4331-5825

[2] Vrije Universiteit Amsterdam, De Boelelaan 1111, 1081 HV Amsterdam, The Netherlands, p.lago@vu.nl, https://orcid.org/0000-0002-2234-0845

Digital sustainability can be considered along four core dimensions: technical, social, economic, and environmental [La15], see Table 1. At its core, a Data Management Maturity Assessment (DMMA) is a systematic process used to evaluate an organisation's DM capabilities against established best practices. Although existing DMMA models, such as DAMA DMMA [DA24] and DCAM [ED26] address various aspects of DM and data governance, we observe a notable gap in research on assessing DM strategies that minimise data volumes and reduce energy consumption while addressing digital sustainability. Our study aims to fill this gap by introducing a model to assess sustainability-aware data DM strategies that address these challenges.

Maturity models describe a progression through levels of process characteristics. By understanding these characteristics, an organisation can evaluate its maturity level and create a plan to improve its capabilities[DA24]. A typical *staged* maturity model, such as the DAMA DMMA, assesses how organisations manage information assets across progressive levels [DA24]. They can be used to evaluate overall DM efforts, or can be used to focus on a single knowledge area or a single process [DA24]. A commonality of DMMA models is the measurement of current and future state [Be21]. Our novel contribution to sustainability-aware DM is using the four core digital sustainability dimensions to define goals, along with CSFs, KPIs, metrics, and measurements to reveal different levels of maturity and effectiveness in reducing data volumes and energy consumption. Our proposed Sustainability-aware Data Management Maturity Assessment model for minimising Data Volumes and Energy consumption (SusDMMA-DVE) can function as a stand-alone or as a supplement to widely used DM frameworks. This is achieved by linking our model to common areas of knowledge in DM, such as Data Governance, Data Architecture, Data Storage and Operations, Data Security, and Data Quality.

Our work in progress builds on previous research by Fellinger and Lago [FL25] that identified nine types of sustainability-aware DM strategies that are applicable in the cloud, data centres, and edge/fog. It also includes our previous research on factors that support or hinder the deployment of sustainability-aware DM strategies [Fe26]. Both studies inform the interviews and case studies that underpin the development of the proposed SusDMMA-DVE model.

A goal is defined as a long-term outcome that teams use to define strategies [FFL25]. Their CSFs are elements that determine long-term success, acting as a focal point for resource allocation, strategy execution, and performance measurement [OC25]. KPIs are the metrics used to measure performance according to the CSFs [OC25]. Based on Lago et al. [La15], DAMA DM-BOK [DA24], ISO25010:2023 [In23], Fellinger [Fe26], and the Cloud Sovereignty Framework of the European Commission [Co25], Table 1 defines the four core dimensions, goals, and CSFs of sustainability-aware DM that we will supplement with the findings of the literature study to build our Preliminary model. The KPIs are specific for the chosen DM strategy; they will be measured in the company context.

Our study design aims to produce the following main contributions:

Core dimension	Goal	CSF	KPI
Technical	Long-term sustainability of data systems and their architectures as they evolve while minimising data volumes and reducing energy consumption	Maintaining product performance within set time and throughput parameters while using resources such as CPU, memory, storage, and bandwidth efficiently along the data pipeline	In company context
Economic	Reducing costs and generating business value with data, ensuring data quality throughout its lifecycle while minimising data volumes and reducing energy consumption.	Lowering operational costs through energy efficiency and reducing data volumes, improving data quality, and maintaining agreed service levels.	In company context
Social	Adhering to laws and data governance principles while minimising data volumes and reducing energy consumption.	Attention to behaviour, risk control, legal compliance (GDPR), security while minimising data volumes and reducing energy use along with sovereignty of data assets and AI services.	In company context
Environmental	Addressing the ecological footprint of data systems while minimising data volumes and reducing energy consumption.	Reducing energy consumption, lowering carbon dioxide emissions, minimising data collection, data storage, and data transmissions	In company context

Tab. 1: Core digital sustainability dimensions, goals, critical success factors (CSFs) and key performance indicators (KPIs).

- A model to assess the maturity and effectiveness of DM strategies that minimise data volumes and reduce energy consumption while balancing the four core dimensions of digital sustainability.
- Our proposed model can function as a stand-alone or as a supplement to other main DM frameworks.

Paper structure Section 2 covers the background and related work. Section 3 details the design of our mixed-method study. Section 4 outlines potential validity threats. Section 5 states the conclusion.

2 Related work

Belghith et al. [Be21] reviewed 20 maturity models related to DM and data governance. Through a comparative analysis of the main concepts and features of these models, a meta model was developed that can serve as a tool for researchers to position future maturity models. However, information on older DMMA models and outdated references highlight the need for ongoing research in this area. Mettler [Me11] outlines the typical phases involved in the development and application of a maturity model from a design science research (DSR) perspective. Using artificial and naturalistic evaluations, we incorporate DSR in a mixed-method study design to define, iteratively refine, and validate our proposed model during development. Lautenschutz et al. [La18] reviewed green ICT maturity models to improve the social or environmental impact of ICT. They applied the knowledge gathered to extend an existing maturity model with the intention of applying it during an ICT sustainability audit in higher education. Their proposal covers a broad range of topics related to energy and material efficiency. In contrast, our research focuses on reducing data volumes and energy consumption. Pörtner et al. [Pö25] researched the maturity levels of data-related challenges during the transition to a data-driven organisation. They state that existing maturity models for digital transformation, data management, and data-driven organisations lack a “comprehensive, industry-agnostic, and practically validated approach to addressing industry challenges”. Corallo et al. [Co23] researched measuring the level of maturity in big data management in manufacturing companies to assess their current status and define future actions in five areas: organisation, DM, data analysis, infrastructure, governance & security. Their DM area has three sub-dimensions: data source, data processing, and data storage, reflecting the data pipeline architecture. However, environmental digital sustainability concerns are not addressed. Yan et al. [YJ26] provides a critical evaluation of 28 data management capability maturity models published over the past decade. The authors highlight several gaps, including an often insufficient coverage of the entire data lifecycle, particularly in relation to data deletion. The Cloud Sovereignty Framework of the European Commission [Co25] aims to protect and ensure control over data and AI services within the European Union (EU). While the EU Cloud Sovereignty Framework is designed for procurement, it may influence DM decision-making within the EU in the years to come. Building on the findings of previous studies, our research aims to address key DM issues related to the four core dimensions of digital sustainability.

3 Study design

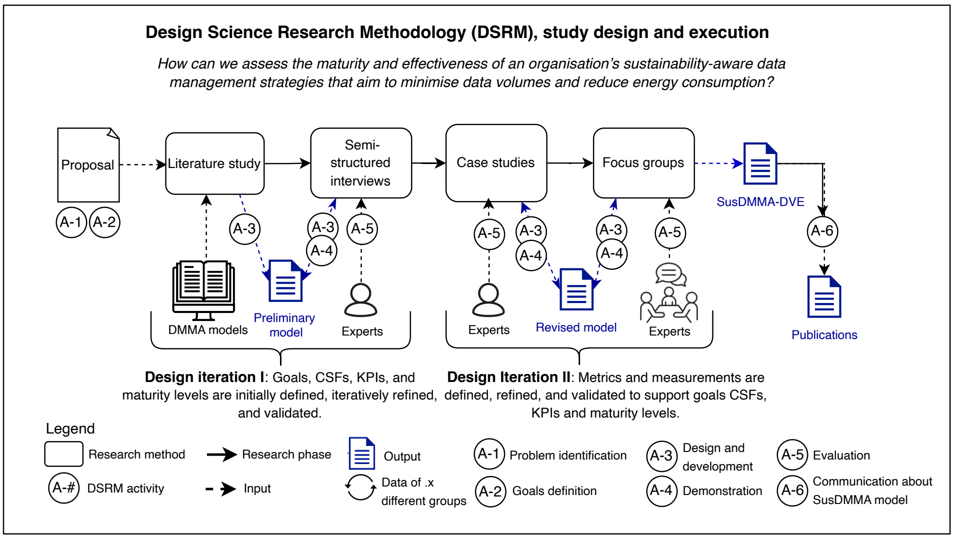

Fig. 1: Study design and execution

In this section, we describe the design of our mixed-method study to develop our proposed SusDMMA-DVE model (see Figure 1). This includes the goal and research question, the collection of data through the literature study, interviews, focus groups, and data analysis. To guide the iterative development of our SusDMMA-DVE model, we use the Design Science Research Methodology (DSRM) in information systems as proposed by Peffers et al. [Pe07]. As shown in Figure 1, the stage of problem identification is covered in this proposal along with the definition of goals. Our proposed SusDMMA-DVE model evolves between two design iterations. During Design Iteration I, the Preliminary model and its goals, CSFs, KPIs, and initial maturity levels are defined through a literature study, then refined and validated with experts. A self-assessment questionnaire is delivered to evaluate sustainability-aware DM strategies aimed at reducing data volumes and energy consumption. Design Iteration II involves the iterative development of a toolkit of metrics and measurement methods through case studies to support goals, CSFs, KPIs, and maturity levels.

In conducting our literature study, we follow the systematic review search methods for grey literature outlined by Godin et al. [Go15]. The guidelines by Wohlin et al. [Wo24] are used to inform the design of the semi-structured interviews and case studies. For the design of the focus groups, we use the guidelines provided by Kontio et al. [KBL08].

3.1 Research goals and research question

The Goal Question Metric (GQM) strategy [BCR94] is used to formulate our research goal: **to analyse** goals, success factors, KPIs, metrics and measurements of DM strategies that focus on minimising data volumes and reducing energy consumption **for the purpose of** understanding their maturity **with respect to** technical, economic, social, and environmental goals **from the point of view of** researchers and practitioners **in the context of** an organisation.

To accomplish the research goal, the study addresses the following RQ:

How can we assess the maturity and effectiveness of an organisation's sustainability-aware DM strategies that aim to minimise data volumes and reduce energy consumption?

3.2 Literature study

To identify current literature on the rapidly evolving field of DMMA models from their developers, accredited partners, users, or critics, we intend to conduct a grey literature search. For a manageable number of search results and to minimise bias from a short time frame while reducing the risk of analysing outdated information, we propose a five-year period for our grey literature search. Grey literature searches can introduce bias in the results due to automated relevance ratings. To mitigate this threat, we will supplement the grey literature with related work (see Section 2), and use the widely recognised DAMA DMMA [DA24] as a reference point.

Identification: we perform our searches using regular Google, employing the search terms "data management maturity" to identify relevant websites that publish documentation on DMMA models. A filter will be applied to the search to capture the results of the last five years. The first ten pages of the search results are captured using the Google Search Results Scraper extension for Google Chrome [Go99]. In this step, the websites are identified and their URL will be entered into an Excel spreadsheet. Next, each of these websites will be manually searched to remove off-topic data, and to identify potentially relevant information on DMMA. The remaining potentially relevant records continue to the screening and selection process. During this step, the name of the website, DMMA model, URL, and year are recorded in an Excel spreadsheet.

Screening and selection process: our review begins with reading the available abstracts, executive summary, table of contents, or introductions to assess relevance to the research objectives using inclusion and exclusion criteria.

Inclusion criteria: I1: Topic: DMMA models I2: Language: written in English I3: Current: the content on DMMA models must have been developed, maintained, or revised within the past five years. I4: Accessible: DMMA documentation must be accessible to researchers. I5: Detail: DMMA documentation should provide details on goals, the knowledge or capability areas assessed, specify progressive levels, metrics, and the assessment process.

Exclusion criteria: E1: Multimedia content: we limit our research to text-based sources. E2: Duplicates: these are removed to avoid invalid conclusions or results. E3: Specific focus: models that focus on assessing a specific capability area, such as data analytics, governance, or security, are excluded.

The full texts of the items that will go to the second stage are then examined. If there is any uncertainty about an item's eligibility, the reviewer will decide to proceed with further screening. The second author will review the selection using a random sample, ensuring that the set criteria were applied objectively and correctly by the first author.

Data collection process and synthesis of results: the remaining records are analysed using thematic analysis [BC06]. Our goal is to extract a superset of features that can be used to assess capabilities related to minimising data volumes and reducing energy consumption toward building the Preliminary version of our proposed SusDMMA-DVE model.

Results: the results of the literature study and our previous research on sustainability-aware DM strategies [FL25], and their supporting and hindering factors [Fe26], inform the first design iteration of the Preliminary model.

3.3 Interviews

The Preliminary model will be used to evaluate the qualitative data from the expert interviews and will be iteratively updated based on the feedback from the interviewees (i. e. expert evaluation). For these interviews, we select experts from companies that are members of the Dutch National Coalition of Sustainable Digitalisation. We interview those who play key roles related to DM and DMMA, such as IT managers, domain experts, data architects, data engineers, and enterprise architects, based on availability (convenience sampling). We plan to use *semi-structured interviews* to balance structured data collection with the flexibility to explore participant insights in depth using the 5W (Who, What, When, Where, Why) and 1H (How) method. The list of DMMA interview topics and questions will be finalised after completing the literature study and creating the Preliminary model.

Informed by our previous work [Fe26; FL25], the literature discussed in the Introduction Section (1) and the Related Work Section (2), we propose to address the following topics for our semi-structured interviews: *Vision and business goals:* To gain an understanding of what our DMMA must support, we begin our inquiry with eliciting the *vision and business goals* in relation to the four core digital sustainability dimensions' goals and CSFs. *Scoping:* After an overview of the situation is made, we determine with the participants the scope of the DMMA: general initiatives, specific branches or departments, knowledge areas, processes. *Detail DM strategies:* To get to the core of the DMMA, we ask whether any DM strategies related to data volume minimisation or energy-savings are applied [FL25]. In addition, we ask for any other sustainability-aware DM efforts that may be relevant. *Data pipeline:* To identify possible overhead in data lineage, we will place the identified DM strategies along the data pipeline; data source, a data processing or data transformation stage, and a data destination or data storage location [HP]. *Factors that support and hinder:* To enrich the Preliminary model with contextual information, we ask about factors that support or hinder the achievement of the goals and CSFs of the four core dimensions of digital sustainability 1.

Data analysis: To identify, analyse and report patterns (themes) within the transcripts from the interviews, we use thematic analysis [BC06] to further refine the output of Design Iteration I. The results are then plotted as capability levels on a radar chart to visually compare them with goals and ambitions across various dimensions.

3.4 Case studies

Design iteration II contains case studies; empirical methods aimed at investigating contemporary phenomena in their context [RH08]. To build the Revised model with participants, we will use case studies to collect quantitative data on effectiveness by measuring KPIs before and after deploying sustainability-aware DM strategies. We propose the following steps, which will be further detailed during the research: **Define metrics:** First, we define metrics to measure the current state of DM strategies relating to the four dimensions' goals, CSFs, and KPIs. We also agree with the participants on the desired levels that will be compared to the actual levels after the measurements. **Measurements:** Participants perform measurements of DM strategies and their KPIs according to agreed and set methods to ensure internal validity. **Analysing data:** Excel is used to record and analyse the quantitative data on KPIs to substantiate our findings. Our goal is to expose current strengths and opportunities for improvement, rather than solving problems. **Data visualisation:** The metrics are plotted as proficiency levels on a radar chart to visually compare the actual maturity with the desired levels and plan improvements to the effectiveness of current DM strategies.

3.5 Focus groups

For the focus groups, we invite experts who play key roles in DM or DMMA (expert sampling). Through the focus group sessions, we aim to further improve our Revised model and its practical application. We will address the suitability of the model for operational use, possible implementation challenges, and suggestions to improve its user-friendly deployment [KBL08]. Data will be captured through observer notes and affinity maps from the sessions. To mitigate validity risks from group dynamics, moderator bias, and contextual factors, we will carefully select participants, train the moderator, and incorporate focus groups into a multi-method research design. After processing the input from the focus groups, design iteration II is completed.

3.6 SusDMMA-DVE

Our proposed SusDMMA-DVE model aims to integrate qualitative and quantitative evidence to determine levels of maturity and effectiveness to reduce data volumes and energy consumption. It consists of a self-assessment questionnaire, descriptions of maturity levels, and a toolkit of metrics and measurement methods to systematically support the findings. We intend to provide two levels of detail to accommodate both managerial overview and specialist needs. To ensure that the proposed model is applicable across multiple types of organisation, we focus on the characteristics of the data involved, rather than the specific industries of the case studies, such as education, banking, retail, or healthcare.

4 Threats to validity

In this section, we discuss the most prominent threats to the validity of our study and how they are mitigated in the study design. We follow the classification of validity threats as discussed by Wohlin et al. [Wo24], based on the work of Cook and Campbell [CC79]. **Construct Validity:** to mitigate the risk of “Inadequate Preoperational Explication of Constructs” [CC79], we use clear definitions of goals, CSFs, and measurement subjects throughout our research. **Internal Validity:** to ensure that any changes are solely due to DM strategies that reduce data volumes and energy consumption, we will clearly define metrics and measurement methods with the participants in advance and calibrate these during our research. **External Validity:** in our interviews and case studies, we use convenience sampling, which limits the ability to draw statistically representative samples. To enhance the generalisability of our findings, we will employ common DM knowledge areas throughout the data pipeline architecture. **Reliability:** to mitigate researcher bias, we will follow a systematic protocol for our literature study, interviews, case studies, and focus groups. We will also provide a public replication package to enable verification of our findings.

5 Conclusion

Our proposed mixed-method study design aims to create the SusDMMA-DVE model, which allows professionals to assess sustainability-aware DM efforts aimed at minimising data volumes and reducing energy consumption. Through gathering qualitative and quantitative evidence, the model can help identify issues such as excessive data collection, processing, and storage hotspots, ultimately reducing resource use and operational costs while meeting business needs.

References

[BC06] Braun, V.; Clarke, V.: Using thematic analysis in psychology. Qualitative Research in Psychology, 3(2), pp. 77–101, 2006.

[BCR94] Basili, V. R.; Caldiera, G.; Rombach, D. H.: The Goal Question Metric Approach. John Wiley & Sons, 1994.

[Be21] Belghith, O. et al.: A Survey of Maturity Models in Data Management. In: 2021 IEEE 12th International Conference on Mechanical and Intelligent Manufacturing Technologies (ICMIMT). Pp. 298–309, 2021.

[Ca24] Castro, V. et al.: Digital data demand and renewable energy limits: Forecasting the impacts on global electricity supply and sustainability. Energy Policy, vol. 195, p. 114404, 2024.

[CC79] Cook, T. D.; Campbell, D. T.: Quasi-experimentation. Houghton Mifflin, Boston, MA, 1979.

[Co23] Corallo, A. et al.: Evaluating maturity level of big data management and analytics in industrial companies. Technological Forecasting and Social Change 196, p. 122826, 2023.

[Co25] Commission, E.: Cloud Sovereignty Framework Version 1.2.1 – Oct. 2025, tech. rep., European Commission Directorate-General for Digital Services, 2025.

[DA24] DAMA: DAMA-DMBOK2 Revision: Data Management Body of Knowledge. DAMA International, 2024.

[ED26] EDM Association: DCAM Framework, Available Online: https://edmcouncil.org/frameworks/dcam/ (Accessed 11th March 2026), 2026.

[Fe26] Fellinger, E.: Assessing the factors that support and hinder sustainability-aware data management strategies. In: Joint Proceedings of ICT4S 2026 Doctoral Symposium, Demonstrations & Posters Track and Workshops. CEUR Workshop Proceedings, in press, CEUR-WS.org, 2026.

[FFL25] Fatima, I.; Funke, M.; Lago, P.: Providing Guidance to Software Practitioners: A Framework for Creating Key Performance Indicators. IEEE Software 42 (4), pp. 68–78, 2025.

[FL25] Fellinger, E.; Lago, P.: Sustainability-driven Data Management Strategies. English, ACM Computing Surveys, (under review), 2025, https://hdl.handle.net/1871.1/639baf52-e8c7-4946-89ea-55a6021913b3.

[Go15] Godin, K. et al.: Applying systematic review search methods to the grey literature: a case study examining guidelines for school-based breakfast programs in Canada. en, Syst. Rev. 4 (1), p. 138, 2015.

[Go99] Google: Google Search Results Scraper, (Accessed May, 1 2026), 9999, https : / / chromewebstore . google . com / detail / google - search - results - scr / cmdkhofenchpdlkbpkheddfgfdmmobcc?hl=en&pli=1.

[HP] HP: Data pipelines, (Accessed May, 1 2026), https://www.hpe.com/nl/en/what-is/data-pipelines.html.

[In23] International Organization for Standardization: ISO/IEC 25010:2023 Systems and software engineering — Systems and software Quality Requirements and Evaluation (SQuaRE) — Product quality model, Available Online: https://www.iso.org/standard/78176.html, 2023.

[KBL08] Kontio, J.; Bragge, J.; Lehtola, L.: The Focus Group Method as an Empirical Tool in Software Engineering. In (Shull, F.; Singer, J.; Sjøberg, D. I. K., eds.): Guide to Advanced Empirical Software Engineering. Springer London, London, pp. 93–116, 2008.

[La15] Lago, P. et al.: Framing sustainability as a property of software quality. Commun. ACM 58 (10), pp. 70–78, 2015.

[La18] Lautenschutz, D. et al.: A Comparative Analysis of Green ICT Maturity Models. In (Penzenstadler, B. et al., eds.): ICT4S2018. 5th International Conference on Information and Communication Technology for Sustainability. Vol. 52. EPiC Series in Computing, EasyChair, pp. 153–167, 2018.

[Me11] Mettler, T.: Maturity assessment models: a design science research approach. Int. J. Soc. Syst. Sci. 3 (1/2), p. 81, 2011.

[OC25] O'Connor, S.: Critical success factors explained: a complete 2026 strategy guide, Available Online: https://monday.com/blog/project-management/critical-success-factors/ (Accessed 9th March 2026), 2025.

[Pe07] Peffers, K. et al.: A Design Science Research Methodology for Information Systems Research. Journal of Management Information Systems 24 (3), pp. 45–77, 2007.

[Pö25] Pörtner, L. et al.: Data Management Maturity Model—Process Dimensions and Capabilities to Leverage Data-Driven Organizations Towards Industry 5.0. Applied System Innovation 8 (2), 2025.

[RH08] Runeson, P.; Höst, M.: Guidelines for conducting and reporting case study research in software engineering. Empirical Software Engineering 14 (2), pp. 131–164, 2008.

[Ve16] Veritas: The Databerg Report:see what others don't identify the value, risk and cost of your data, tech. rep., Veritas, 2016, https://www.veritas.com/content/dam/Veritas/docs/reports/veritas-strike-global-report_a4-sdc2.pdf.

[Wa23] Watts, M.: Data Waste Index, Available Online: "https://www.netapp.com/pdf.html?item=/media/83312-data_waste_research_report_final_for_submission.pdf", 2023.

[Wo24] Wohlin, C. et al.: Experimentation in software engineering. Springer, Berlin, Germany, 2024.

[YJ26] Yan, S.; Jones, J.: Bridging the Gaps: A Comprehensive Review of Data Management Capability Maturity Models in the Digital Era. IEEE Transactions on Big Data 12 (2), pp. 307–320, 2026.